

# *A Neural Network Based Approach for Reliability Analysis of UAV Flapping Wing*

Yaojia Han  
School of Aeronautics  
Northwestern Polytechnical  
University  
Xi'an, China  
hyjnwpu@163.com

Jieyu Jia  
School of Aeronautics  
Northwestern Polytechnical  
University  
Xi'an, China  
Jiajieyu@mail.nwpu.edu.cn

Weimin Cui  
School of Aeronautics  
Northwestern Polytechnical  
University  
Xi'an, China  
cuiwm@nwpu.edu.cn

Fangyi Wan  
School of Aeronautics  
Northwestern Polytechnical  
University  
Xi'an, China  
fwan@nwpu.edu.cn

Yugang Zhang  
School of Aeronautics  
Northwestern Polytechnical  
University  
Xi'an, China  
zhangyugang@nwpu.edu.cn

**Abstract**—During the flight experiment of the flapping-wing UAV, the hinged position of the crank and connecting rod of the mechanism is worn very seriously, and there is significant non-uniform wear phenomenon, which quickly leads to the unbalance of the mechanism's movement until the mechanism parts are broken and the movement is terminated. In order to solve the reliability problem caused by the wear of the rotating pair in the complex mechanism, this paper constructs the functional functions of the flapping-wing drive mechanism, investigates the random distribution of the factors affecting the reliability of the mechanism, and establishes the training samples of the proxy model. Based on the BP neural network method, the agent model under 50 different cycles is obtained, and the wear depth distribution of the mechanism in each period is obtained successively, according to which the reliability evolution of the mechanism is analyzed. The research content of this paper is of great significance for further reliability analysis of the driving mechanism of the flapping-wing UAV and providing suggestions for improved design.

**Keywords**—driving mechanism, reliability analysis, BP neural network method

## I. INTRODUCTION

The driving mechanism of a flapping-wing aircraft [1] is one of the key factors affecting the flight performance of the entire aircraft. It is an energy conversion device for a small flapping-wing aircraft, which realizes flapping flight by converting electrical energy into mechanical energy of flapping-wing reciprocating flapping. In the actual flight process, the bushing of the drive mechanism wears severely, which will quickly cause the hinge hole at the wear area to deform [2], and also cause other subsequent failures, destroying the originally designed motion law. When the wear further intensifies [3], the connecting rod will shake seriously [4]. Therefore, this study is of great significance to the reliability analysis of the flapping-wing drive mechanism considering wear.

Due to the rapid increase in the amount of calculation for engineering structural problems, the model technology is gradually being applied in various research projects. At present, the most widely used proxy model approximation techniques include response surface models [5], Kriging interpolation [6,7] and artificial neural networks [8]. In the early 1990s, the approximate function of artificial neural networks (especially feedback neural networks) extended to a large number of researches on structural optimization design problems. Adeli [9] et al. used artificial neural network models for the first time in the field of structural engineering design, providing an intelligent solution for structural design and analysis; In 1993, Berke [10] et al. found that neural networks can be used to optimize the design of civil and aerospace engineering Structural components.

Based on the analysis of flight experiments, the non-uniform wear of the hinged position of the crank connecting rod of the flapping wing drive mechanism is the most important failure mode [11]. The connecting rod at the hinge is exposed to uneven stress from the bearing for a long time, resulting in fatigue wear and severe deformation leading to failure [12]. Fig.1 shows the experimental results of the wear of the bushing at the hinge of the flapping wing mechanism.

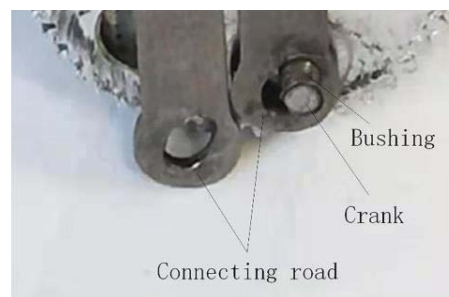


Fig.1. Experimental results of bush wear

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There are certain assembly tolerances in the installation process of mechanical products, and in the case of a certain cost, the parts themselves also have certain manufacturing tolerances. Under the action of both will make it easier for the flapping wing drive mechanism in the manufacture after put into use, hinge clearance position has certain deviation, motion and kinematic changes in form. Finally, the flight stability of flapping wing aircraft is affected, and the fatigue life of bushing is uneven. Therefore, this paper takes the material and size parameters of the bushing as random variables to analyze the reliability of the flapping-wing drive mechanism.

## II. FUNCTION ANALYSIS OF FLAPPING WING DRIVE MECHANISM

### A. Definition of Institutional Reliability

In the failure mode analysis, if the strength limit and the allowable value of performance parameters are regarded as generalized strength, and the structural stress of components and the performance parameters of the mechanism are regarded as generalized stress, then the corresponding failure reason of the mechanism is that the generalized stress is greater than the generalized strength. Based on the generalized stress-strength interference theory, its reliability can be expressed as:

$$P_r = P(R \geq S) = P(G(R, S) \geq 0) \quad (1)$$

In the formula:  $R$  is the generalized strength and  $S$  is the generalized stress.

### B. Function of flapping wing drive mechanism

The failure of flapping wing drive mechanism means that the bushing fracture occurs because the depth of non-uniform wear exceeds the allowable wear depth. Generally speaking, the greater the wear of the mechanical components, the greater the probability of failure. In the analysis and calculation of wear reliability, the allowable wear amount that is usually used as the failure criterion is generally the linear wear amount. The corresponding linear wear can be obtained by subtracting the initial gap with the new gap after the wear of two matching surfaces after movement.

The method for determining the allowable wear amount of the hinge is based on the literature, when the nominal diameter is represented by  $D$ ,  $U_{\max} = 0.009 \sim 0.018D$  is taken under the condition of general accuracy requirements. When there is no specific data, the recommended parameters for hinge bearing  $U_{\max}$  are:  $\mu_{U_{\max}} \approx 0.01D$ ,  $\sigma_{U_{\max}} \approx 0.0002 \sim 0.0005D$ . The calculated bushing diameter at the crank connecting rod of the flapping wing drive mechanism is 3mm, and the wear allowance of the corresponding bushing is 0.03mm. Since the initial radius inside the bushing is 1.002mm, once the maximum wear depth of the bushing exceeds 1.032mm, the mechanism movement will fail. The function of the flapping wing drive mechanism is defined as follows:

$$g = H_c - H_{\max} \quad (2)$$

Among them,  $H_c$  is the allowable amount of bush wear, and  $H_{\max}$  is the maximum wear depth of the bush.

When  $g > 0$ , the flapping-wing mechanism operates normally; when  $g < 0$ , the flapping-wing mechanism fails;

when  $g = 0$ , the mechanism is in a critical state, and here,  $H_{\max} = 1.032\text{mm}$ .

## III. THE INFLUENCING FACTORS OF MECHANISM RELIABILITY AND ITS RANDOM ANALYSIS

### A. Randomness of influencing factors

In the deterministic design of the mechanism, a reasonably designed mechanism should have a higher generalized strength and a certain safety margin than the generalized stress. In addition to being affected by many factors, both the generalized strength and generalized stress of the mechanism have a certain degree of randomness. Therefore, even in a reasonably designed mechanism, the generalized stress may be higher than the generalized strength. In addition, the degradation of material properties and the accumulation of component damage during use will also lead to a decrease in generalized strength and an increase in generalized stress, thus increasing the probability of failure. It can be seen that the root causes of mechanism failure include the randomness of reliability influencing factors and the accumulation of component damage.

As shown in the figure below, the design with safety margin may still fail due to random influencing factors, and when the safety margin is equal, the reliability decreases as the randomness increases.

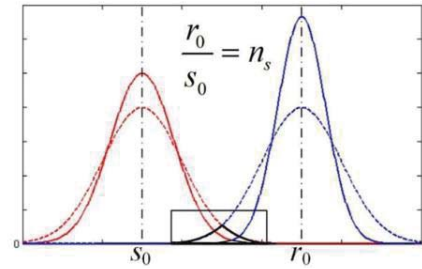


Fig.2. Random influencing factors affect the reliability of the organization

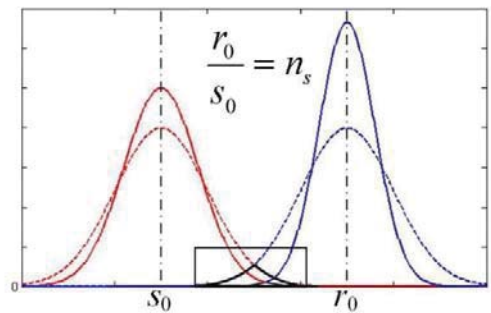


Fig.3. The detailed diagram in the box shown in Fig.2

### B. Analysis of Influencing Factors of Uneven Wear Failure

In the reliability analysis of the flapping wing drive mechanism, six influencing factors are mainly considered: the initial inner radius of the bushing, the bushing wear rate, the friction coefficient, the thickness of the bushing, the Young's

modulus and the Poisson's ratio. In fact, due to the existence of manufacturing and assembly errors, these six influencing factors are not equal to the theoretical design values, but have a certain degree of randomness. Changes in these factors will affect the performance and reliability of the mechanism. Therefore, in the reliability analysis of the drive mechanism, the above 6 parameters are treated as random variables, and the maximum wear depth on the bush after the mechanism runs 50,000 cycles (hereinafter referred to as the "maximum wear depth after 50,000 cycles") as the response Variable Y.

The normal distribution is a general distribution. However, in practice, the value of the variable does not go from negative infinity to positive infinity, so this article assumes that the six variables all obey the truncated normal distribution. Among them, the mean value of the variable is obtained by consulting relevant information. It is reasonable to assume that the coefficient of variation is 0.1 to obtain the value of the corresponding standard deviation. The distribution parameters of random variables are shown in Table I.

TABLE I. RANDOM VARIABLE DISTRIBUTION PARAMETER TABLE

| Serial number | Variable/Unit             | Variable meaning         | Distributed         | Mean  | Lower limit | Upper limit | Standard deviation |
|---------------|---------------------------|--------------------------|---------------------|-------|-------------|-------------|--------------------|
| 1             | $Z_0/(\text{mm})$         | initial inner radius     | normal distribution | 1.002 | 1.000       | 1.004       | 0.1002             |
| 2             | wearrate $/(mm^3/Nmm)$    | bushing wear rate        | normal distribution | 5.05  | 5           | 5.1         | 0.505              |
| 3             | frico/(-)                 | friction coefficient     | normal distribution | 0.1   | 0.09        | 0.11        | 0.01               |
| 4             | thick/(mm)                | thickness of the bushing | normal distribution | 1     | 0.995       | 1.005       | 0.1                |
| 5             | $E/(10^{11} \cdot N/m^2)$ | Young's modulus          | normal distribution | 1.24  | 1.18        | 1.3         | 0.124              |
| 6             | Lamda/(-)                 | Poisson's ratio          | normal distribution | 0.32  | 0.3         | 0.34        | 0.032              |

#### IV. CONSTRUCTION AND VERIFICATION BASED ON PROXY MODEL

Failure function of flapping wing drive mechanism is difficult to clearly display, and the simulation model requires about 450s to calculate a sample point. If the reliability evaluation is directly performed by Monte Carlo, its huge computational cost is unbearable. Therefore, the proxy model is used to evaluate the reliability of the flapping-wing drive mechanism. Considering that the proxy model constructed by the polynomial response surface and Kriging model can only simulate a single output, this article has 36 outputs, in order to avoid multiple calculations, the BP neural network method that can achieve multiple inputs and multiple outputs is used for analysis.

##### A. Construct samples

First, sample the 6 random variables according to the ideal normal distribution, inspect and remove the samples that fall outside the truncation interval, and continue sampling until the required number of samples is drawn, which is 1000. Subsequently, these variables are brought into the established mechanism motion model one by one to simulate and calculate in LMS Virtual.Lab to obtain the corresponding output variable values. Organize these input variables and their response values into a folder to facilitate subsequent data analysis. Each

sample takes about 450s, so it took 125 hours to build the training sample of the proxy model, which requires a lot of work.

##### B. Proxy model construction and verification

In order to prevent the neural network from overfitting the training data and deviating from other data that has not participated in the training, the established training samples are allocated to training, validation and test according to the ratio of 60%, 20% and 20%. Among them, only training data is used for training, while the other two parts of data are only used for testing.

Taking 1000 cycles as the equivalent wear cycle T, a proxy model is established for the mechanism's operation of 50,000 cycles, that is 50 cycles, to analyze the changes in the reliability of the mechanism. Next, we select five equally spaced periods to construct the proxy model. This article only shows the results of the fifth wear period's proxy model in detail.

##### (1) 50th wear cycle

After calculation, when the number of hidden layer units of the network is set to 10, the transfer function is set to the sigmoid function, the initial weight matrix is a small random amount generated by the system, and the rest of the settings are in accordance with the system default values when the BP neural network training situation is the best. The calculation results are as follows: Fig.4 shows that the system is iteratively calculated 9 times in total, the calculation time is only 1 second, the network output error is  $1.27 \times 10^{-8}$ , and the iterative gradient is  $7.09 \times 10^{-8}$ .

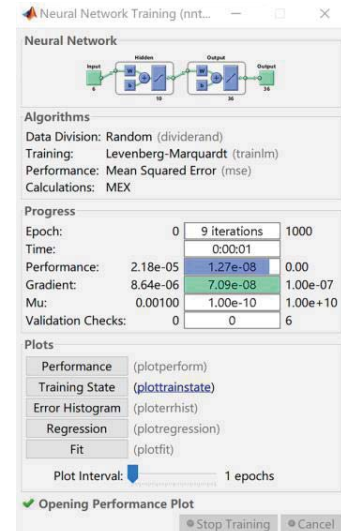


Fig.4. Neural network related parameters

MSE (Mean Squared Error) is the Mean Squared Error, which displays the degree of data change and is a convenient way to measure the average error. In the MATLAB neural network, it is a measure of the training situation of the neural network in each generation. The smaller the value, the more accurate the calculation is and the more ideal the result is. Fig.5 shows that the training result is the best when iterating for the

9th time, and the mean square error (MSE) is only  $1.3484 \times 10^{-8}$ .

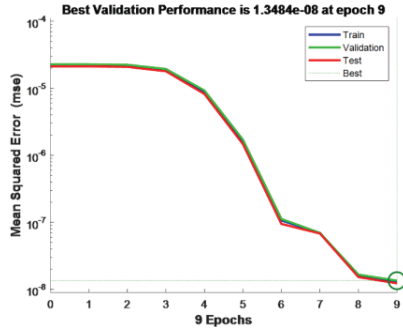


Fig.5. Neural network mean square error (force)

In order to prevent the neural network from over-fitting the training data and deviating from other data that did not participate in the training, the Matlab system automatically divides the input samples into three parts - Training, Validation and Test. Only training data is used for training, and the other two parts of data are only used for testing. Fig.6 is a regression diagram of the network output and the target value of the training and test results. The R value indicates the degree of correlation between the network output and the target expected value, and an R of 1 indicates complete correlation, namely the best fitting situation. The figure shows that the overall fit of the neural network is 0.99994. This result shows that the learning of BP neural network is good enough to simulate the functions of flapping wing driving mechanism.

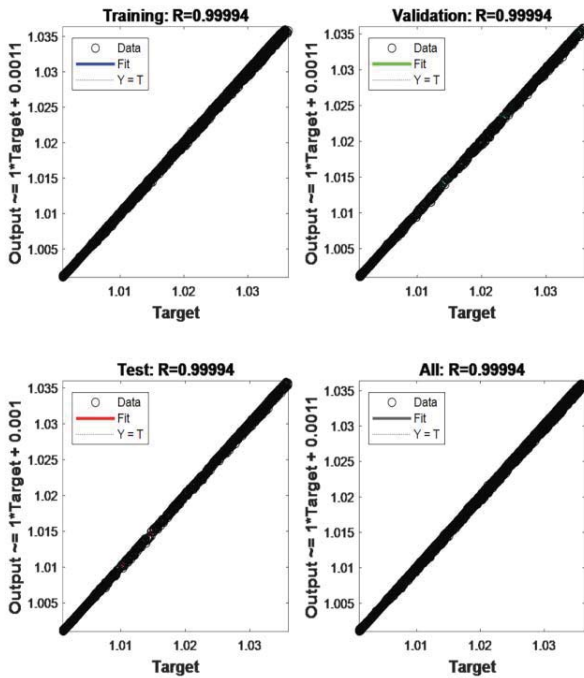


Fig.6. Neural network sample training results (force)

## V. RELIABILITY ANALYSIS OF FLAPPING WING DRIVE MECHANISM CONSIDERING WEAR

According to the basic process of reliability analysis, 50 proxy models are constructed respectively, and  $10^6$  samples are drawn separately according to the Monte Carlo method. The maximum wear depth corresponding to different wear cycles of the lower flapping wing drive mechanism of each sample was calculated, and draw the result histogram as shown. It can be seen that as the wear cycle increases, the distribution of the maximum wear depth of the bushing gradually changes, and the average value continues to increase.

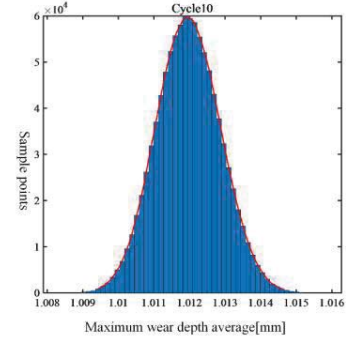


Fig.7. Maximum wear distribution of 10 cycle

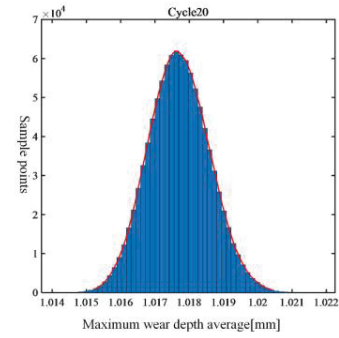


Fig.8. Maximum wear distribution of 20 cycle

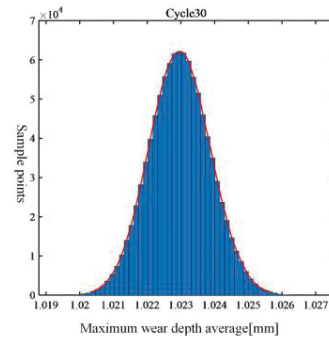


Fig.9. Maximum wear distribution of 30 cycle



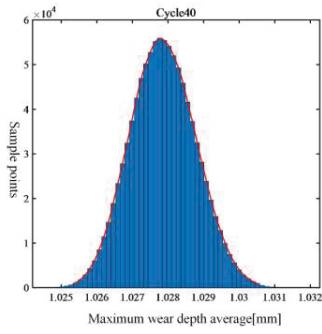


Fig.10. Maximum wear distribution of 40 cycle

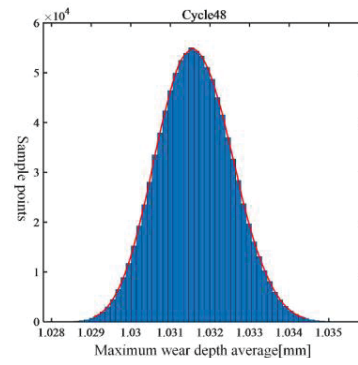


Fig.14. Maximum wear distribution of 48 cycle

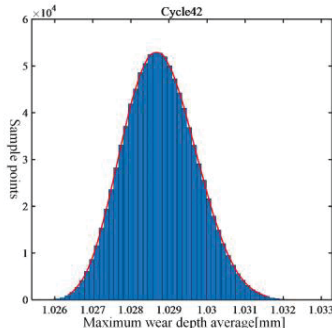


Fig.11. Maximum wear distribution of 42 cycle

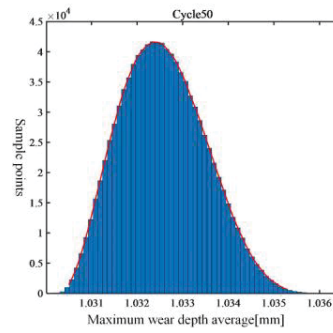


Fig.15. Maximum wear distribution of 50 cycle

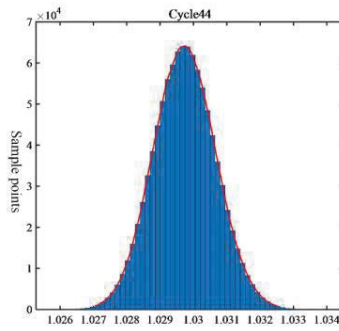


Fig.12. Maximum wear distribution of 44 cycle

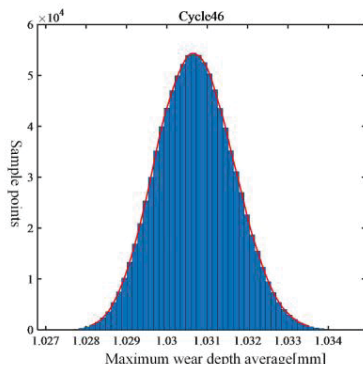


Fig.13. Maximum wear distribution of 46 cycle

Based on this, the curve of the average value of the maximum wear depth of the bushing with the wear cycle is drawn, as shown in Fig.15. It can be seen from the figure that the bush wear is in a rapid wear period before the 5th wear cycle, then the wear rate decreases and is in a stable wear period, until the 40th wear cycle begins to accelerate wear. This wear law also accords with the rule of bathtub curve.

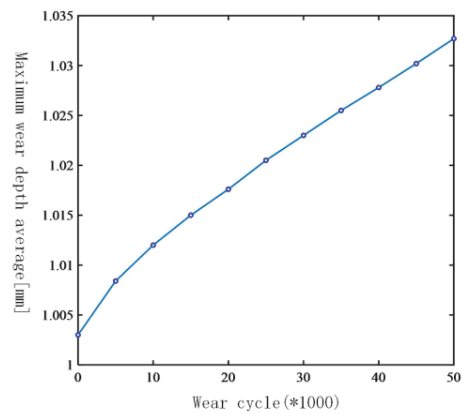


Fig.16. The average change curve of the maximum wear depth

At the same time, the reliability of the mechanism under different periods is calculated according to the function. The variation of mechanism reliability is shown in Table II .

TABLE II. STATISTICAL TABLE OF INSTITUTIONAL RELIABILITY CHANGES

| Serial number | Wear cycle | Reliability |
|---------------|------------|-------------|
| 0             | 0          | 1           |
| 1             | .....      | .....       |
| 2             | 40         | 1           |
| 3             | 41         | 1           |
| 4             | 42         | 0.9998      |
| 5             | 43         | 0.9985      |
| 6             | 44         | 0.9908      |
| 7             | 45         | 0.9653      |
| 8             | 46         | 0.9217      |
| 9             | 47         | 0.8012      |
| 10            | 48         | 0.6401      |
| 11            | 49         | 0.4573      |
| 12            | 50         | 0.2596      |

The reliability evolution curve of the corresponding flapping-wing drive mechanism is shown in Fig.16. Since the changes are concentrated between the 40th to 50th wear cycles, a partial enlarged view is drawn, as shown in Fig.17.

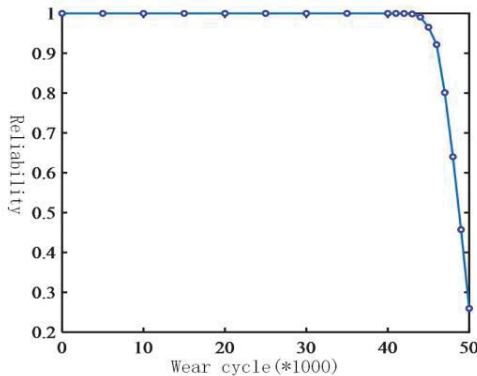


Fig. 17. Reliability evolution curve of flapping-wing drive mechanism

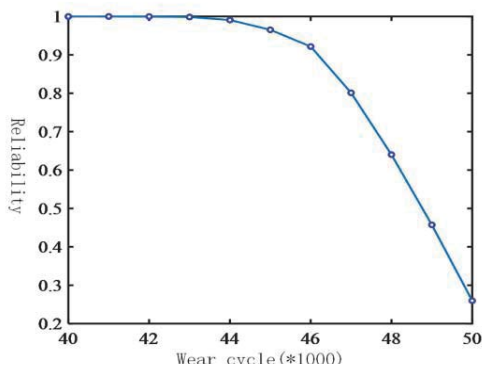


Fig. 18. Partial enlarged view of reliability evolution curve

As can be seen from the above figure, the flapping wing drive mechanism always maintained high reliability before the 42nd equivalent wear cycle, and then began to slowly decrease until the 46th equivalent wear cycle, when the reliability began to decline sharply.

## VI. CONCLUSION

In this paper, the reliability analysis and calculation of the flapping-wing drive mechanism considering uneven wear are carried out. Combined with the mechanism reliability evaluation method, the maximum wear depth of the flapping-wing drive mechanism exceeds the allowable value as the failure mode, and the initial factors and use factors of the flapping-wing drive mechanism are selected to conduct a reliability analysis of the flapping-wing drive mechanism considering wear. The functional function of the mechanism was established by referring to relevant manuals, and 6 main influencing factors and their random distribution were considered at the same time. By calculating the corresponding agency response values, a training sample for the agent model is established. Based on the BP neural network method, the agency's wear depth proxy model in 50 different cycles is further obtained. Subsequently, a large number of samples were drawn according to the Monte Carlo method to obtain the wear depth distribution curve of the mechanism in different cycles. According to this, on the one hand, the change curve of the maximum wear depth on the bushing is obtained, and the wear rule is analyzed; On the other hand, the reliability evolution curve of the mechanism is obtained.

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